

● HARD

● INFORMATION AND IDEAS

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13

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

Almost all works of fiction contain references to the progression of time, including the time of day when events in a story take place. In a 2020 study, Allen Kim, Charuta Pethe, and Steven Skiena claim that an observable pattern in such references reflects a shift in human behavior prompted by the spread of electric lighting in the late nineteenth century. The researchers drew this conclusion from an analysis of more than 50,000 novels spanning many centuries and cultures, using software to recognize and tally both specific time references—that is, clock phrases, such as 7 a.m. or 2:30 p.m.—and implied ones, such as mentions of meals typically associated with a particular time of day.

Which finding from the study, if true, would most directly support the researchers' conclusion?

- A. Novels published after the year 1800 include the clock phrase 10 a.m. less often than novels published before the year 1800 do.
- B. Novels published after 1880 contain significantly more references to activities occurring after 10 p.m. than do novels from earlier periods.
- C. Among novels published in the nineteenth century, implied time references become steadily more common than clock phrases as publication dates approach 1900.
- D. The time references of noon (12 p.m.) and midnight (12 a.m.) are used with roughly the same frequency in the novels.

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers' reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____ blank

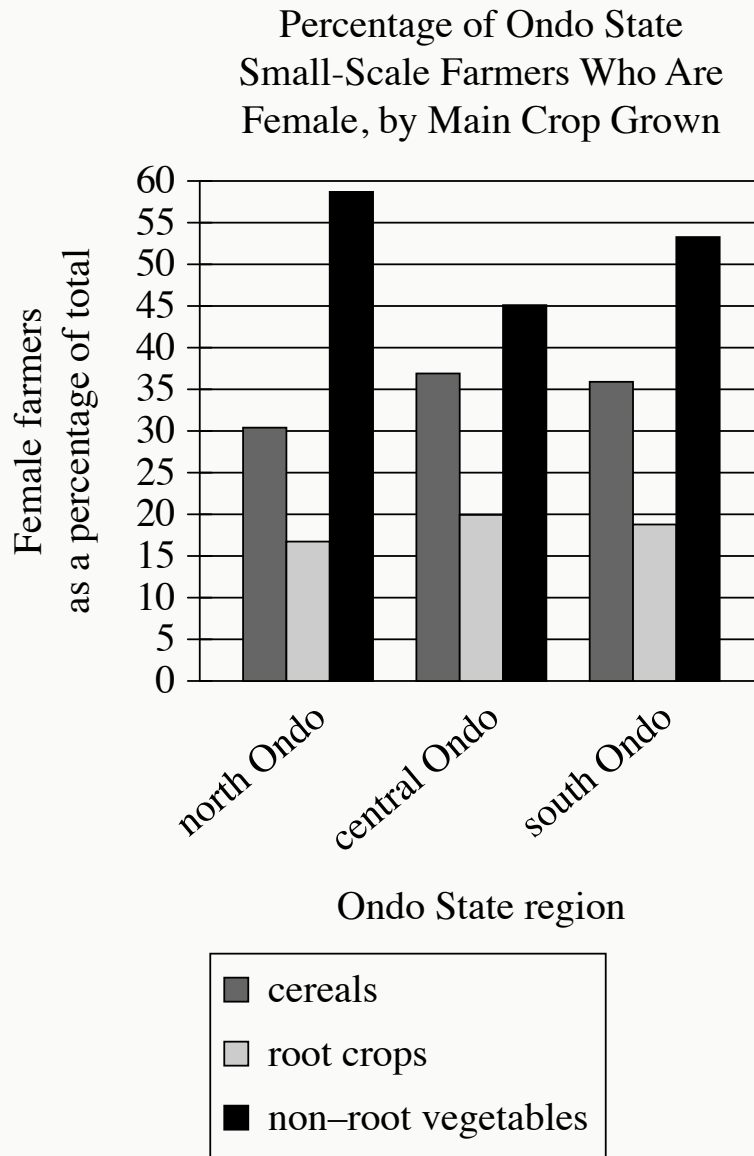
Which choice most logically completes the text?

- A. find the disruptions more irritating as time went on.
- B. rate the listening experience as more negative than those whose listening experience was uninterrupted.
- C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.
- D. perceive the volume of the construction noise as growing softer over time.

Algae living within the tissues of corals play a critical role in keeping corals, and the marine ecosystems they are part of, thriving. Some coral species appear brown in color when healthy due to the algae colonies living in their tissues. In the event of an environmental stressor, the algae can die or be expelled, causing the corals to appear white. To recover the algae, the bleached corals then begin to produce bright colors, which block intense sunlight, encouraging the light-sensitive algae to recolonize the corals.

What does the text most strongly suggest about corals that produce bright colors?

- A. These corals have likely been subjected to stressful environmental conditions.
- B. These corals are likely more vulnerable to exposure from intense sunlight than white corals are.
- C. These corals have likely recovered from an environmental event without the assistance of algae colonies.
- D. These corals are more likely to survive without algae colonies than brown corals are.



- For each data category, the following bars are shown:
 - cereals
 - root crops
 - non-root vegetables
- The data for the 3 categories are as follows:
 - north Ondo:
 - cereals: 30.4%

- root crops: 16.73%
- non-root vegetables: 58.69%
- central Ondo:
 - cereals: 36.9%
 - root crops: 19.93%
 - non-root vegetables: 45.1%
- south Ondo:
 - cereals: 35.9%
 - root crops: 18.78%
 - non-root vegetables: 53.27%

Geographer Adebayo Oluwole Eludoyin and his colleagues surveyed small-scale farmers in three locations in Ondo State, Nigeria—which has mountainous terrain in the north, an urbanized center, and coastal terrain in the south—to learn more about their practices, like the types of crops they mainly cultivated. In some regions, female farmers were found to be especially prominent in the cultivation of specific types of crops and even constituted the majority of farmers who cultivated those crops; for instance, _____blank

Which choice most effectively uses data from the graph to complete the example?

- A. most of the farmers who mainly cultivated cereals and most of the farmers who mainly cultivated non-root vegetables in south Ondo were women.
- B. more women in central Ondo mainly cultivated root crops than mainly cultivated cereals.
- C. most of the farmers who mainly cultivated non-root vegetables in north and south Ondo were women.
- D. a relatively equal proportion of women across the three regions of Ondo mainly cultivated cereals.

Laura Mulvey has theorized that in narrative film, shots issuing from a protagonist's point of view compel viewers to identify with the character. Such identification is heightened by "invisible editing," or editing so inconspicuous that it renders cuts between shots almost unnoticeable. Conversely, Mulvey proposes that conspicuous editing or an absence of point-of-view shots would induce a more critical stance toward a protagonist. Consider, for example, the attic scene in Alfred Hitchcock's *The Birds*, a conspicuously edited sequence of tens of shots, few of which correspond to the protagonist's point of view. According to Mulvey's logic, this scene should affect viewers by _____ blank

Which choice most logically completes the text?

- A. obscuring their awareness of the high degree of artifice involved in constructing the montage.
- B. lessening their identification with the protagonist, if not alienating them from the character altogether.
- C. compelling them to identify with the film's director, whose proxy is the camera, and not with the protagonist.
- D. diverting their attention away from the film's content and toward its stylistic attributes.

Gorgetts, or necklaces with large pendants, have been part of the ceremonial attire of tribes from the US Southeast for centuries. One of the oldest examples, the Fairfield Gorget, was found in Fairfield, Missouri, in 1958. Its overall design resembles that of other art from the region during the Mississippian period (900–1600 CE). Yet the image on the gorget is of a jaguar—a species whose range doesn’t extend to Missouri. Jaguar images are common in ancient Mexican art, and Mexico lies squarely in the species’ range. Therefore, some scholars argue that long-distance trade in Mexican art objects brought the imagery to Missouri, where a local artist could have adopted it and incorporated it into the Fairfield Gorget.

Which finding, if true, would most strongly support the underlined explanation?

- A. The Fairfield Gorget is dated not to the Mississippian period but instead to the earlier Woodland period, which ended around 900 CE.
- B. The range of the jaguar is shown to have expanded dramatically after the Mississippian period came to a close around 1600 CE.
- C. An ancient Mexican art object is found at a site that dates to the Mississippian period and is close to where the Fairfield Gorget was found.
- D. Certain works by present-day artists from Southeastern tribes reflect outside cultural influences, including contemporary Mexican art.

A heliograph is a semaphore device used for sending optical communications—usually in the form of Morse code—by reflecting flashes of sunlight off a mirror. Heliographs were used for rapid communication across expansive distances for military, surveying, and forestry purposes during the late nineteenth and early twentieth centuries, but they were largely effective only during the daytime, and the range of the device depended on factors such as the opacity of the air and line of sight. Therefore, heliographs were eventually replaced by technology that _____ blank

Which choice most logically completes the text?

- A. worked on similar principles but was easier to produce and maintain.
- B. was not so constrained by environmental circumstances.
- C. could be used for more than military, surveying, or forestry purposes.
- D. enabled communication that didn't require knowledge of Morse code.

A student in a political science course is writing a paper on Aristotle's *The Politics*, in which Aristotle offers his opinion on political instability and gives advice on how constitutions can be preserved. Aristotle observes that different forms of government can fall in different ways—for example, oligarchies might grant power to military leaders during wartime who refuse to relinquish that power during peacetime—but some methods of preserving order apply across all forms of government. The student claims that in particular Aristotle asserts that in a healthy state obedience to law must be as close to absolute as possible and that even minor infractions should not be ignored.

Which quotation from a philosopher's analysis of *The Politics* would best support the student's claim?

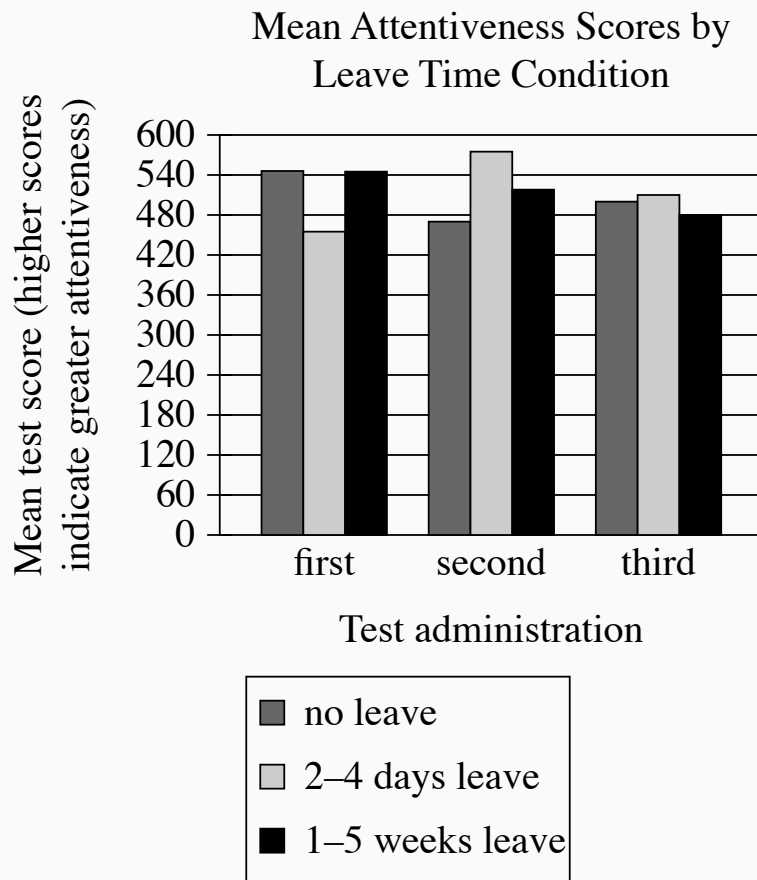
- A. "When constructing his argument regarding the characteristics of a well-functioning government, Aristotle asserts that 'Transgression creeps in unperceived and at last ruins the state,' illustrating this idea with a comparison to frequent small expenditures slowly and almost imperceptibly chipping away at a fortune until it is ultimately depleted."
- B. "When Aristotle writes on the necessity of avoiding corruption in government, he proposes that 'every state should be so administered and so regulated by law that its magistrates cannot possibly make money.' In particular, he thinks oligarchies are particularly susceptible to corruption through bribery."
- C. "When Aristotle considers the health of constitutions, he states that 'Constitutions are preserved when their destroyers are at a distance, and sometimes also because they are near, for the fear of them makes the government keep in hand the constitution.' He holds that rulers who wish to see constitutions preserved must continually remind the populace of the dangers that would result from a constitutional collapse."
- D. "When contrasting different forms of government, Aristotle holds that 'oligarchies may last, not from any inherent stability in such forms of government, but

because the rulers are on good terms both with the unenfranchised and with the governing classes.’ That is, oligarchic leaders who wish to hold on to power will introduce members of disenfranchised classes into government in a participatory role.”

Researchers Suchithra Rajendran and Maximilian Popfinger modeled varying levels of passenger redistribution from short-haul flights (flights of 50 to 210 minutes, from takeoff to landing) to high-speed rail trips. Planes travel faster than trains, but air travel typically requires 3 hours of lead time for security, baggage handling, and boarding that rail travel doesn't, so short-haul routes take similar amounts of time by air and by rail. However, the model suggests that as rail passenger volumes approach current capacity limits, long lead times emerge. Therefore, for rail to remain a viable alternative to short-haul flights, _____ blank

Which choice most logically completes the text?

- A. rail systems should offer fewer long-haul routes and airlines should offer more long-haul routes.
- B. rail systems may need to schedule additional trains for these routes.
- C. security, baggage handling, and boarding procedures used by airlines may need to be implemented for rail systems.
- D. passengers who travel by rail for these routes will need to accept that lead times will be similar to those for air travel.



- For each data category, the following bars are shown:
 - no leave
 - 2-4 days leave
 - 1-5 weeks leave
- The data for the 3 categories are as follows:
 - First test administration:
 - no leave: 546
 - 2-4 days leave: 455
 - 1-5 weeks leave: 545
 - Second test administration:
 - no leave: 470
 - 2-4 days leave: 575
 - 1-5 weeks leave: 518

- Third test administration:
 - no leave: 500
 - 2–4 days leave: 510
 - 1–5 weeks leave: 480

To investigate potential cognitive benefits of taking leave from work, psychologist Jan Packer and colleagues conducted a six-month study of Australian university staff members who took no leave from work during the study, took 2–4 days of leave, or took 1–5 weeks of leave. Tests of attentiveness were administered to participants three times during the study: at random for the no-leave staff, and for the rest, one week before their leave, one week following their return to work, and one week after the second test administration. After analyzing the results, the researchers concluded that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Which choice best describes data from the graph that support the researchers' conclusion?

- A. In the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, but in the third test administration, those who took no leave had higher average scores than those who took 1–5 weeks of leave.
- B. In the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did those who took 1–5 weeks of leave and those who took no leave.
- C. In both the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did participants who took 1–5 weeks of leave.
- D. In the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave.